

D522

Python for IT Automation

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A disk-monitor script for a data center with 400 servers sets usage = 92 and then runs: if usage > 80: print('warn'), elif usage > 90: print('critical'), else: print('ok'). What prints?

- A. ok
- B. critical
- C. warn
- D. warn and then critical

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A script must print critical for usage = 92 and warn for usage = 85. How should the conditions be ordered?

- A. Replace elif with a second if.
- B. Test usage > 90 first, then elif usage > 80.
- C. Change elif usage > 90 to else.
- D. Change > 80 to >= 80 and keep the order.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A script at a university with 30,000 students retries a ping: attempts = 0, then while attempts < 3 it pings once and adds 1 to attempts, stopping early only on success. The network team upgraded switches last month. If every ping fails, how many pings run?

- A. 3
- B. It loops forever, because a failing ping never ends a while loop.
- C. 4
- D. 2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A retry loop must try at most 5 times, and attempts starts at 1 and increases by 1 after each try. Which condition is correct?

- A.** while attempts > 5
- B.** while True
- C.** while attempts < 5
- D.** while attempts <= 5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

An IT technician at a clinic with 9 laptops writes `def free_gb(): print(42)` and then `total = free_gb() + 10`. What happens?

- A.** 42 prints, then a `TypeError` because `None` cannot be added to 10.
- B.** 52 prints.
- C.** total becomes 10.
- D.** total becomes 52.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Which definition makes `total = free_gb() + 10` equal 52?

- A.** `def free_gb(): return 42`
- B.** `def free_gb(): 42`
- C.** `def free_gb(): return '42'`
- D.** `def free_gb(): print(42)`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A log-rotation script for a hosting firm with 1,500 customers opens files with `f = open(path)` and never closes them. After several hours it fails with “too many open files.” The script runs on a server bought in 2021. Which change fixes it best?

- A.** Replace the 2021 server.
- B.** Raise the operating system's open-file limit.
- C.** Call `f.close()` only after a successful read.
- D.** Use with `open(path) as f:`, which closes each file automatically, even on errors.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

An automation script at a school district with 60 buildings takes a hostname typed by help desk staff, builds a command string, and runs it with subprocess and shell=True. The help desk has 7 staff. What is the safer design?

- A. Keep shell=True, because internal staff can always be trusted.
- B. Wrap the call in try/except.
- C. Switch to os.system with the same string.
- D. Validate the hostname, pass the command as a list without shell=True, and check the return code.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A backup script at a bank with 120 branches stops the whole nightly run when one branch server is unreachable. The script was written in 2019. What change makes it resilient?

- A. Retry the unreachable server forever.
- B. Catch the connection error for each server, log it, retry a few times, and continue with the rest.
- C. Wrap everything in a bare except: pass.
- D. Upgrade to a newer Python release because the script dates from 2019.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A script reads the line 'ERROR 2025-06-01 disk07 full' from a storage array with 48 disks and runs parts = line.split(). What is parts[2]?

- A. '48'
- B. '2025-06-01'
- C. 'disk07'
- D. 'R'

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Given line = 'user=admin;role=ops', which expression returns 'ops'?

- A. line.split(';')[1].split('=')[0]
- B. line.split(';')[1].split('=')[1]
- C. line.split('=')[1]
- D. line.split(';')[0].split('=')[1]

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

12.

A new automation script for a manufacturer with 5 plants has the service account password typed directly in the code, which lives in a shared Git repository. The repository has 14 contributors. What is the best fix?

- A. Base64-encode the password in the code.
- B. Cut the repository down to fewer contributors.
- C. Remove the secret from the code, rotate the password, and load it at runtime from an environment variable or secrets manager.
- D. Delete the line but keep the same password.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A disk-monitor script for a data center ~~with 400 servers~~ sets usage = 92 and then runs: if usage > 80: print('warn'), elif usage > 90: print('critical'), else: print('ok'). What prints?

Target word: branch that runs **Decides:** 92 > 80 is the first true test.

A	Inverted operation	else runs only when every test is false.
B	Confusion set	Assumes the most specific branch wins; order wins.
C	Correct	The first true branch runs; the elif is never checked.
D	Adjacent category	That would need two separate if statements.

2. Answer B Concept 1 · transfer

A script must print critical for usage = 92 and warn for usage = 85. How should the conditions be ordered?

Target word: fix the order — reverse **Decides:** The narrower test must come first.

A	Confusion set	92 would print both warn and critical.
B	Correct	92 hits the first branch; 85 falls to the second.
C	Adjacent category	else would catch low values too.
D	Wrong stem number	92 still matches the first branch.

3. Answer A Concept 2

A script at a university ~~with 30,000 students~~ retries a ping: attempts = 0, then while attempts < 3 it pings once and adds 1 to attempts, stopping early only on success. ~~The network team upgraded switches last month.~~ If every ping fails, how many pings run?

Target word: iteration count **Decides:** attempts takes values 0, 1, 2 while < 3.

A	Correct	Runs at 0, 1, and 2; stops when attempts reaches 3.
B	Absolute language	The counter ends it.
C	Confusion set	Counts the final check that ends the loop as a run.
D	Dropped element	Drops the run at attempts = 0.

4. Answer D Concept 2 · transfer

A retry loop must try at most 5 times, and attempts starts at 1 and increases by 1 after each try. Which condition is correct?

Target word: condition for a count — reverse **Decides:** Values 1 through 5 must pass.

A	Inverted operation	Never runs.
B	Adjacent category	No limit unless something else breaks the loop.
C	Dropped element	Runs only 4 times.
D	Correct	Runs for 1, 2, 3, 4, 5.

5. Answer A Concept 3

An IT technician at a clinic ~~with 9 laptops~~ writes `def free_gb(): print(42)` and then `total = free_gb() + 10`. What happens?

Target word: result **Decides:** The function prints but returns None.

A	Correct	With no return, the function gives None.
B	Confusion set	Nothing prints the sum, and the sum fails.
C	Adjacent category	None is not treated as zero.
D	Dropped element	Treats print as if it returned the value.

6. Answer A Concept 3 · transfer

Which definition makes `total = free_gb() + 10` equal 52?

Target word: choose the definition — reverse **Decides:** The function must return the integer 42.

A	Correct	Returns 42, so <code>42 + 10 = 52</code> .
B	Dropped element	No return statement, so None.
C	Adjacent category	A string plus an int raises <code>TypeError</code> .
D	Confusion set	Returns None.

7. Answer D Concept 4

A log-rotation script for a hosting firm ~~with 1,500 customers~~ opens files with `f = open(path)` and never closes them. After several hours it fails with “too many open files.” ~~The script runs on a server bought in 2021.~~ Which change fixes it best?

Target word: fix **Decides:** Files are never closed.

A	Camouflage	The leak follows the script to any server.
B	Adjacent category	Delays the failure without fixing the leak.
C	Dropped element	Leaks the file whenever an error occurs.
D	Correct	The context manager guarantees closing.

8. Answer D Concept 5

An automation script at a school district ~~with 60 buildings~~ takes a hostname typed by help desk staff, builds a command string, and runs it with `subprocess` and `shell=True`. ~~The help desk has 7 staff.~~ What is the safer design?

Target word: safe design **Decides:** User text reaches a shell.

A	Absolute language	Mistakes and compromised accounts happen.
B	Adjacent category	Handles errors but does not stop unsafe input.
C	Confusion set	Also passes the string to a shell.
D	Correct	Keeps user input as data, not shell code.

9. Answer B Concept 6

A backup script at a bank ~~with 120 branches~~ stops the whole nightly run when one branch server is unreachable. ~~The script was written in 2019.~~ What change makes it resilient?

Target word: resilience change **Decides:** One failure halts every branch.

A	Absolute language	Blocks the rest of the run.
B	Correct	Failures are contained, recorded, and bounded.
C	Dropped element	Hides failures with no log or retry.
D	Camouflage	The version is not the cause.

10. Answer C Concept 7

A script reads the line 'ERROR 2025-06-01 disk07 full' ~~from a storage array with 48 disks~~ and runs `parts = line.split()`. What is `parts[2]`?

Target word: list element **Decides:** Zero-based: ERROR, date, disk07, full.

A	Camouflage	The disk count is not in the line.
B	Confusion set	Counts from 1.
C	Correct	Index 2 is the third item.
D	Adjacent category	Indexes the string, not the split list.

11. Answer B Concept 7 · transfer

Given `line = 'user=admin;role=ops'`, which expression returns 'ops'?

Target word: choose the expression — reverse **Decides:** Split on ';', take index 1, split on '=', take index 1.

A	Inverted operation	Returns the key 'role', not the value.
B	Correct	'role=ops' then 'ops'.
C	Dropped element	Returns 'admin;role'; skips the ';' split.
D	Confusion set	Returns 'admin'.

12. Answer C Concept 8

A new automation script for a manufacturer ~~with 5 plants~~ has the service account password typed directly in the code, which lives in a shared Git repository. ~~The repository has 14 contributors.~~ What is the best fix?

Target word: secret-handling fix **Decides:** A secret is hard-coded and shared.

A	Confusion set	Encoding is not encryption; anyone can decode it.
B	Camouflage	The secret is still in the code.
C	Correct	Removes exposure and invalidates the leaked value.
D	Dropped element	The old value stays in Git history.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	3	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	if / elif / else order	Python runs only the first branch whose condition is true; put the most specific test first.	Expecting a later, more specific elif to run.	Overlapping conditions like <code>> 80</code> and <code>> 90</code> .
2	Loop bounds	A while loop runs as long as its condition is true; count the iterations from the start value and the comparison.	Off-by-one counts, or assuming a failing task loops forever.	A counter, a limit, and “how many times?”
3	return vs. print	print shows a value; return hands it back to the caller. A function with no return gives None.	Using a printed value in a calculation.	A function's result used in an expression.
4	Context managers for files	with open(...) as f closes the file automatically, even when errors occur.	Opening files without closing them, or closing only on the success path.	“Too many open files.”
5	Safe subprocess use	Validate input, pass command arguments as a list without shell=True, and check the return code.	Building a shell string from user input because users are trusted.	User-typed text passed into a shell command.
6	Resilient error handling	Catch specific exceptions per task, log them, retry a limited number of times, and continue.	Bare except: pass, or retrying forever.	One failure stops the whole run.
7	Parsing strings	str.split() returns a list; index it from 0, and chain splits for nested delimiters.	Counting from 1, or indexing the string instead of the list.	A log or config line and a split.
8	Handling secrets	Never hard-code credentials; load them at runtime from environment variables or a secrets manager, and rotate any exposed secret.	Encoding the password (e.g., Base64) or deleting the line without rotating.	A password in source code or a shared repository.

Course scope (public description): Python for IT Automation covers the fundamentals of the Python language and its features to control program flow, inform decisions, and automate IT tasks and processes. The course emphasizes a systematic approach to solving problems and the application of programming logic to administer secure, scalable, and resilient IT networks and systems.